

CHAPTER

6.1

TRANSFER FUNCTION

1. The equivalent transfer function of three parallel blocks

$$G_1(s) = \frac{1}{s+1}, G_2(s) = \frac{1}{s+4} \text{ and } G_3(s) = \frac{s+3}{s+5} \text{ is}$$

- (A) $\frac{(s^3 + 10s^2 + 34s + 37)}{(s+1)(s+4)(s+5)}$
 (B) $\frac{(s+3)}{(s+1)(s+4)(s+5)}$
 (C) $\frac{-(s^3 + 10s^2 + 34s + 37)}{(s+1)(s+4)(s+5)}$
 (D) $\frac{-(s+3)}{(s+1)(s+4)(s+5)}$

2. The block having transfer function

$$G_1(s) = \frac{1}{s+2}, G_2(s) = \frac{1}{s+5}, G_3(s) = \frac{s+1}{s+3}$$

are cascaded. The equivalent transfer function is

- (A) $\frac{(s^3 + 10s^2 + 37s^2 + 31)}{(s+2)(s+3)(s+5)}$
 (B) $\frac{s+1}{(s+2)(s+3)(s+5)}$
 (C) $\frac{-(s^3 + 10s^2 + 37s^2 + 31)}{(s+2)(s+3)(s+5)}$
 (D) $\frac{-(s+1)}{(s+2)(s+3)(s+5)}$

3. For a negative feedback system shown in fig. P.6.1.3

$$G(s) = \frac{s+1}{s(s+2)} \text{ and } H(s) = \frac{s+3}{s+4}$$

The equivalent transfer function is

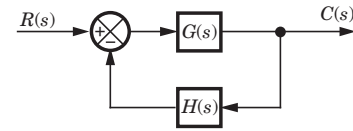


Fig. P.6.1.3

- (A) $\frac{s(s+2)(s+3)}{s^3 + 7s^2 + 12s + 3}$
 (B) $\frac{s(s+2)(s+3)}{s^3 + 5s^2 + 4s - 3}$
 (C) $\frac{(s+1)(s+4)}{s^3 + 7s^2 + 12s + 3}$
 (D) $\frac{(s+1)(s+4)}{s^3 + 5s^2 + 4s - 3}$

4. A feedback control system is shown in fig. P.6.1.4. The transfer function for this system is

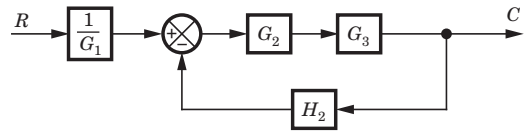


Fig. P.6.1.4

- (A) $\frac{G_1 G_2}{1 + H_1 G_1 G_2 G_3}$
 (B) $\frac{G_2 G_3}{G_1 (1 + H_1 G_2 G_3)}$
 (C) $\frac{G_2 G_3}{1 + H_1 G_1 G_2 G_3}$
 (D) $\frac{G_2 G_3}{G_1 (1 + H_1 G_2 G_3)}$

5. Consider the system shown in fig. P.6.1.5.

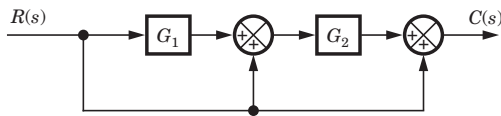
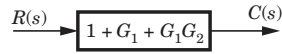


Fig. P.6.1.5

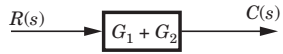
The input output relationship of this system is



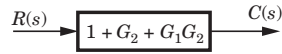
(A)



(B)



(C)



(D)

6. A feedback control system shown in fig. P.6.1.6 is subjected to noise $N(s)$.

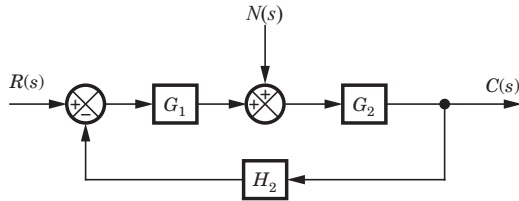


Fig. P.6.1.6

The noise transfer function $\frac{C_N(s)}{N(s)}$ is

(A) $\frac{G_2}{1 + G_1 G_2 H}$

(B) $\frac{G_2}{1 + G_1 H}$

(C) $\frac{G_2}{1 + G_2 H}$

(D) None of the above

7. A system is shown in fig. P.6.1.7. The transfer function for this system is

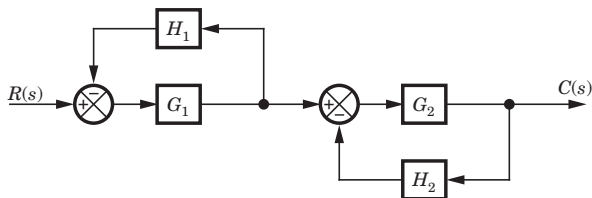


Fig. P.6.1.7

(A) $\frac{G_1 G_2}{1 + G_1 G_1 H_2 + G_2 H_1}$

(B) $\frac{G_1 G_2}{1 + G_1 G_2 + H_1 H_2}$

(C) $\frac{G_1 G_2}{1 - G_1 H_1 - G_2 H_2 + G_1 G_2 H_1 H_2}$

(D) $\frac{G_1 G_2}{1 + G_1 H_1 + G_2 H_2 + G_1 G_2 H_1 H_2}$

8. The closed loop gain of the system shown in fig. P.6.1.8 is

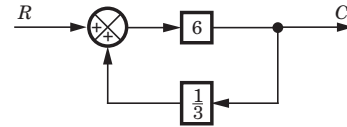


Fig. P.6.1.8

(A) -2

(B) 6

(C) -6

(D) 2

9. The block diagrams shown in fig. P.6.1.9 are equivalent if G is equal to

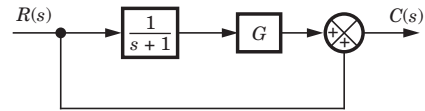
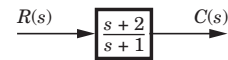


Fig. P.6.1.9

(A) $s + 1$

(B) 2

(C) $s + 2$

(D) 1

10. Consider the systems shown in fig. P.6.1.10. If the forward path gain is reduced by 10% in each system, then the variation in C_1 and C_2 will be respectively

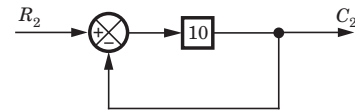
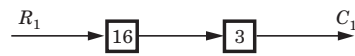


Fig. P.6.1.10

(A) 10% and 1%

(B) 2% and 10%

(C) 10% and 0%

(D) 5% and 1%

11. The transfer function $\frac{C}{R}$ of the system shown in the fig. P.6.1.11 is

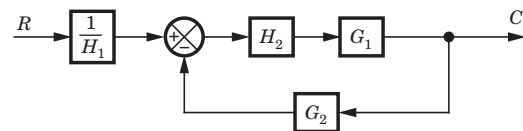


Fig. P.6.1.11

- (A) $\frac{G_1 H_2}{H_1(1 + G_1 G_2 H_2)}$ (B) $\frac{G_1 G_2 H_2}{H_1(1 + G_1 G_2 H_2)}$
 (C) $\frac{G_2 G_1}{1 + H_1 H_2 G_1 G_2}$ (D) $\frac{G_1 G_2}{H_1(1 + G_1 G_2 H_2)}$

12. In the signal flow graph shown in fig. P.6.1.12 the sum of loop gain of non-touching loops is

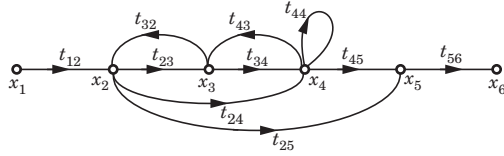


Fig. P.6.1.12

- (A) $t_{32}t_{23} + t_{44}$ (B) $t_{23}t_{32} + t_{34}t_{43}$
 (C) $t_{24}t_{43}t_{32} + t_{44}$ (D) $t_{23}t_{32} + t_{34}t_{43} + t_{44}$

13. For the SFG shown in fig. P.6.1.14 the graph determinant Δ is

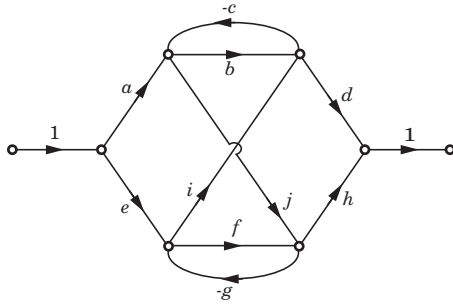


Fig. P.6.1.13

- (A) $1 - bc - fg - bcfg + cigj$
 (B) $1 - bc - fg - cigj + bcfg$
 (C) $1 + bc + fg + cigj - bcfg$
 (D) $1 + bc + fg + bcfg - cigj$

14. The sum of the gains of the feedback paths in the signal flow graph shown in fig. P.6.1.14 is

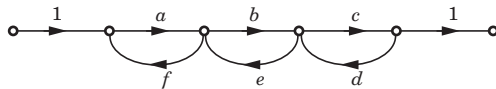


Fig. P.6.1.14

- (A) $af + be + cd + abef + bcde$
 (B) $af + be + cd$
 (C) $af + be + cd + abef + abcdef$
 (D) $af + be + cd + cbef + bcde + abcdef$

15. A closed-loop system is shown in fig. P.6.1.15. The noise transfer function $C_n(s)/N(s)$ is approximately

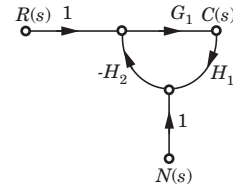


Fig. P.6.1.15

- (A) $\frac{1}{G_1(s)H_1(s)}$ For $|G_1(s)H_1(s)H_2(s)| \ll 1$
 (B) $\frac{1}{-H_1(s)}$ For $|G_1(s)H_1(s)H_2(s)| \gg 1$
 (C) $\frac{1}{H_1(s)H_2(s)}$ For $|G_1(s)H_1(s)H_2(s)| \gg 1$
 (D) $\frac{1}{G_1(s)H_1(s)H_2(s)}$ For $|G_1(s)H_1(s)H_2(s)| \ll 1$

16. The overall transfer function $\frac{C}{R}$ of the system shown in fig. P.6.1.16 will be

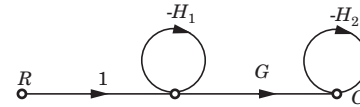


Fig. P.6.1.16

- (A) G (B) $\frac{G}{1 + H_2}$
 (C) $\frac{G}{(1 + H_1)(1 + H_2)}$ (D) $\frac{G}{1 + H_1 + H_2}$

17. Consider the signal flow graphs shown in fig. P.6.1.17. The transfer 2 is of the graph

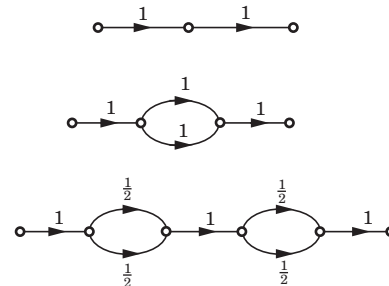
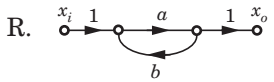
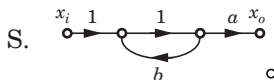
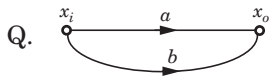
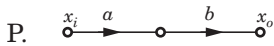


Fig. P.6.1.17

- (A) a (B) b
 (C) b and c (D) a, b and c

18. Consider the List I and List II

List I
(Signal Flow Graph)



List II
(Transfer Function)

1. $a + b$

2. ab

3. $\frac{a}{(1 - ab)}$

4. $\frac{a}{1 - b}$

The correct match is

	P	Q	R	S
(A)	2	1	3	4
(B)	2	1	4	3
(C)	1	2	4	3
(D)	1	2	3	4

19. For the signal flow graph shown in fig. P6.1.19 an equivalent graph is

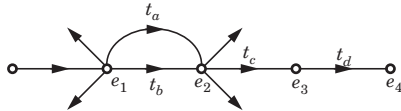
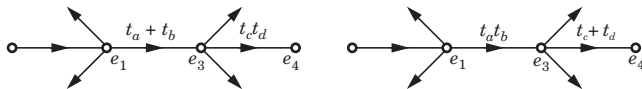
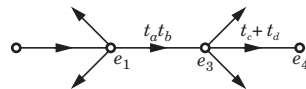


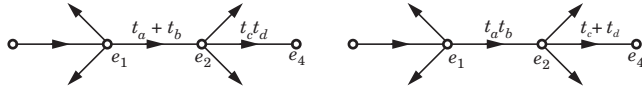
Fig. P.6.1.19



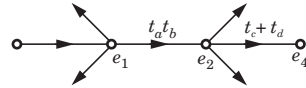
(A)



(B)



(C)



(D)

20. Consider the block diagram shown in figure P.6.1.20

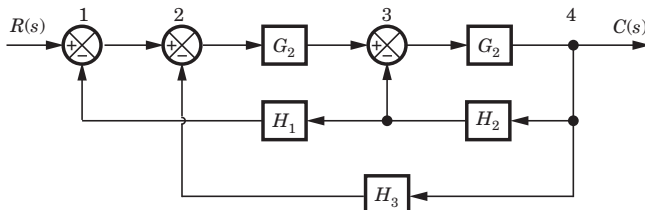
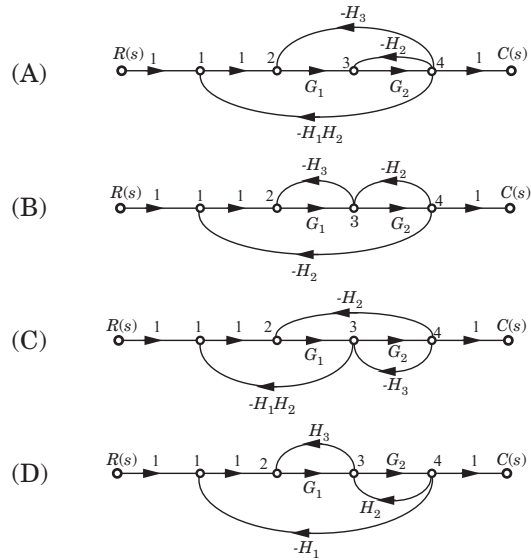


Fig. P.6.1.20

For this system the signal flow graph is



21. The block diagram of a system is shown in fig. P.6.1.21. The closed loop transfer function of this system is

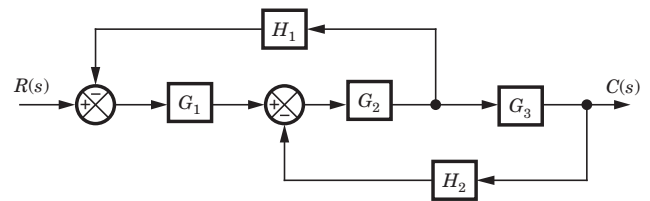


Fig. P.6.1.21

- (A) $\frac{G_1 G_2 G_3}{1 + G_1 G_2 G_3 H_1}$
- (B) $\frac{G_1 G_2 G_3}{1 + G_1 G_2 G_3 H_1 H_2}$
- (C) $\frac{G_1 G_2 G_3}{1 + G_1 G_2 H_1 + G_2 G_3 H_2}$
- (D) $\frac{G_1 G_2 G_3}{1 + G_1 G_2 H_1 + G_1 G_3 H_2 + G_2 G_3 H_1}$

22. For the system shown in fig. P.6.1.22 transfer function $C(s)/R(s)$ is

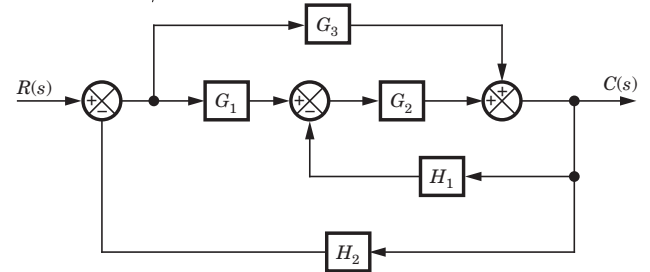


Fig. P.6.1.22

$$(D) \frac{G_3}{1 - H_1 G_2 - H_2 G_3 - G_1 G_2 H_2}$$

$$(D) \frac{G_1 G_3 + G_2 G_3}{1 + G_1 G_3 H_1 + G_2 G_3 H_1 + G_4}$$
$$(B) \ G_6[G_2 + G_3 + G_5(G_3 + G_4)]$$

(C) $G_6[G_1 + G_2 + G_3(G_4 + G_5)]$

(D) None of the above

29. For the block diagram shown in fig. P.6.1.29 the transfer function $C(s)/R(s)$ is

(A) $\frac{50(s-2)}{s^3 + s^2 + 150s - 100}$ (B) $\frac{50(s-2)}{s^3 + s^2 + 150s}$

(C) $\frac{50s}{s^3 + s^2 + 150s - 100}$ (D) $\frac{50}{s^2 + s + 150}$

30. For the SFG shown in fig. P.6.1.30 the transfer function $\frac{C}{R}$ is

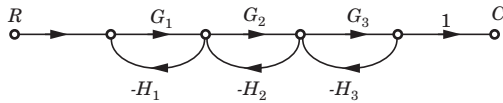


Fig. P.6.1.30

(A) $\frac{G_1 + G_2 + G_3}{1 + G_1H_1 + G_2H_2 + G_3H_3}$

(B) $\frac{G_1 + G_2 + G_3}{1 + G_1H_1 + G_2H_2 + G_3H_3 + G_1G_3H_1H_3}$

(C) $\frac{G_1G_2G_3}{1 + G_1H_1 + G_2H_2 + G_3H_3}$

(D) $\frac{G_1G_2G_3}{1 + G_1H_1 + G_2H_2 + G_3H_3 + G_1G_3H_1H_3}$

31. Consider the SFG shown in fig. P.6.1.31. The Δ for this graph is

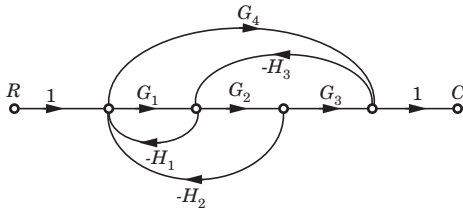


Fig. P.6.1.31

(A) $1 + G_1H_1 + G_2G_3H_3 + G_1G_3H_2$

(B) $1 + G_1H_1 - G_2G_3H_3 - G_1G_3H_3 + G_2G_4H_2H_3$

(C) $1 + G_1H_1 + G_2G_3H_3 + G_1G_3H_3 - G_2G_4H_2H_3$

(D) $1 + G_1H_1 + G_2G_3H_3 + G_1G_3H_3 + G_2G_4H_2H_3$

32. The transfer function of the system shown in fig. P.6.1.32 is

(A) $\frac{G_2G_3 + G_1G_3}{1 - G_3H_1 + G_2G_3}$ (B) $\frac{G_2G_3 + G_1G_3}{1 + G_3H_1 - G_2G_3}$

(C) $\frac{G_2G_3 + G_1G_3}{1 + G_3H_1 + G_2G_3}$ (D) $\frac{G_2G_3 + G_1G_3}{1 - G_3H_1 - G_2G_3}$

33. The closed loop transfer function of the system

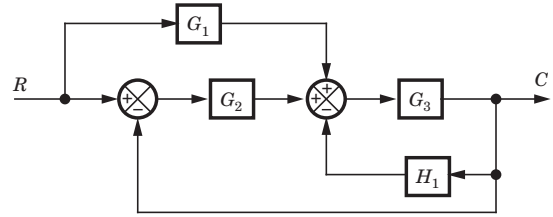


Fig. P.6.1.32

shown in fig. P.6.1.33 is

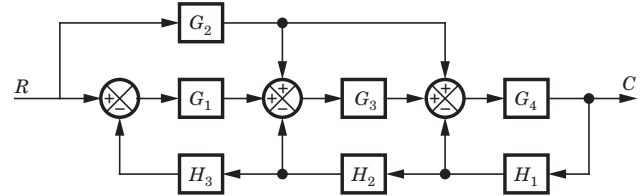


Fig. P.6.1.33

(A) $\frac{G_1G_2G_3 + G_2G_3G_4 + G_1G_4}{1 + G_1G_3G_4H_1H_2H_3 + G_2G_4H_1H_2 + G_4H_1}$

(B) $\frac{G_2G_4 + G_1G_2G_3}{1 + G_1G_3H_1H_2H_3 + G_4H_1 + G_3G_4H_1H_2}$

(C) $\frac{G_1G_3G_4 + G_2G_4}{1 + G_3G_4H_1H_2 + G_4H_1 + G_1G_3H_3H_2}$

(D) $\frac{G_1G_3G_4 + G_2G_3G_4 + G_2G_4}{1 + G_1G_3G_4H_1H_2H_3 + G_3G_4H_1H_2 + G_4H_1}$

Statement for Q.34-37:

A block diagram of feedback control system is shown in fig. P.6.1.34-37

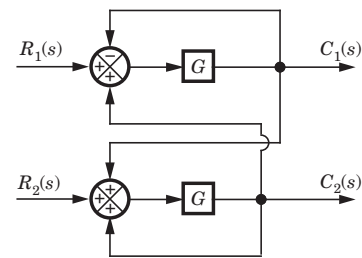


Fig. P.6.1.34-37

34. The transfer function $\frac{C_1}{R_1} \Big|_{R_2=0}$ is

(A) $\frac{G}{1 - 2G^2}$ (B) $\frac{G(1 - G)}{1 - 2G^2}$

(C) $\frac{G(1 - 2G)}{1 - G^2}$ (D) $\frac{G}{1 - G^2}$

35. The transfer function $\left. \frac{C_1}{R_2} \right|_{R_1=0}$ is

- (A) $\frac{G}{1-2G^2}$ (B) $\frac{G}{1-G^2}$
 (C) $\frac{G^2}{1-2G^2}$ (D) $\frac{G^2}{1-G^2}$

36. The transfer function $\left. \frac{C_2}{R_1} \right|_{R_2=0}$ is

- (A) $\frac{G(1+G)}{1-2G^2}$ (B) $\frac{G^2}{1-2G^2}$
 (C) $\frac{G^2}{1-G^2}$ (D) $\frac{G}{1-G^2}$

37. The transfer function $\left. \frac{C_2}{R_2} \right|_{R_1=0}$ is

- (A) $\frac{G(1+G)}{1-2G^2}$ (B) $\frac{G}{1-2G^2}$
 (C) $\frac{G}{1+G}$ (D) $\frac{G}{1-G^2}$

Statement for Q.38–39:

A signal flow graph is shown in fig. P.6.1.38–39.

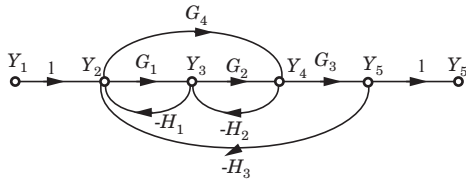


Fig. P.6.1.38-39

38. The transfer function $\frac{Y_2}{Y_1}$ is

- (A) $\frac{1}{\Delta}$ (B) $\frac{1+G_2H_2}{\Delta}$
 (C) $\frac{G_1G_2G_3}{\Delta}$ (D) None of the above

39. The transfer function $\frac{Y_5}{Y_2}$ is

- (A) $\frac{G_1G_2G_3+G_4G_3}{\Delta}$ (B) $G_1G_2G_3+G_4G_3$
 (C) $\frac{G_1G_2G_3+G_4G_3}{G_1G_2G_3}$ (D) $\frac{G_1G_2G_3+G_4G_3}{1+G_2H_2}$

Statement for Q.40–41:

A block diagram is shown in fig. P6.1.40–41.

40. The transfer function for this system is

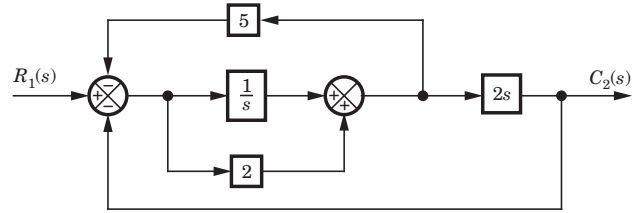


Fig. P.6.1.40-41

- (A) $\frac{2s(2s+1)}{2s^2+3s+5}$ (B) $\frac{2s(2s+1)}{2s^2+13s+5}$
 (C) $\frac{2s(2s+1)}{4s^2+13s+5}$ (D) $\frac{2s(2s+1)}{4s^2+3s+5}$

41. The pole of this system are

- (A) $-0.75 \pm j1.39$ (B) $-0.41, -6.09$
 (C) $-0.5, -1.67$ (D) $-0.25 \pm j0.88$

SOLUTIONS

1. (A) $G_e(s) = G_1(s) + G_2(s) + G_3(s)$

$$= \frac{1}{(s+1)} + \frac{1}{(s+4)} + \frac{s+3}{(s+5)}$$

$$= \frac{s^2 + 9s + 20 + s^2 + 6s + 5 + s^3 + 5s^2 + 4s + 3s^2 + 15s + 12}{(s+1)(s+4)(s+5)}$$

$$= \frac{s^3 + 10s^2 + 34s + 37}{(s+1)(s+4)(s+5)}$$

2. (B) $G_e(s) = G_1(s)G_2(s)G_3(s) = \frac{(s+1)}{(s+2)(s+5)(s+3)}$

3. (C) $\frac{C(s)}{R(s)} = \frac{G(s)}{1 + H(s)G(s)}$

$$= \frac{\frac{s+1}{s(s+2)}}{1 + \frac{(s+3)(s+1)}{(s+4)s(s+2)}} = \frac{(s+1)(s+4)}{s^3 + 7s^2 + 12s + 3}$$

4. (B) Multiply G_2 and G_3 and apply feedback formula and then again multiply with $\frac{1}{G_1}$.

$$T(s) = \frac{G_2G_3}{G_1(1 + G_2G_3H_1)}$$

5. (D) $T(s) = G_2(1 + G_1) + 1 = 1 + G_1 + G_1G_2$

6. (A) Open-loop gain = G_2

Feed back gain = HG_1 $T_N(s) = \frac{G_2}{1 + G_1G_2H}$

7. (D) Apply the feedback formula to both loop and then multiply

$$T(s) = \left(\frac{G_1}{1 + G_1H_1} \right) \left(\frac{G_2}{1 + G_2H_2} \right)$$

$$= \frac{G_1G_2}{1 + G_1H_1 + G_2H_2 + G_1G_2H_1H_2}$$

8. (C) For positive feedback $\frac{C}{R} = \frac{6}{1 - \frac{6 \times 1}{3}} = -6$

9. (D) For system (b) closed loop transfer function

$$\frac{G}{s+1} + 1 = \frac{G+s+1}{s+1}, \quad \frac{G+s+1}{s+1} = \frac{s+2}{s+1}, \quad \text{Hence } G = 1$$

10. (A) In open loop system change will be 10% in C_1 also but in closed loop system change will be less

$$C_2 = \frac{10}{10+1} = \frac{10}{11}, \quad C'_2 = \frac{9}{9+1} = \frac{9}{10}, \quad C_2 \text{ is reduced by } 1\%.$$

11. (A) Apply the feedback formula and then multiply by $\frac{1}{H_1}$,

$$\frac{C}{R} = \frac{(H_2G_1) \left(\frac{1}{H_1} \right)}{1 + H_2G_1G_2} = \frac{H_2G_1}{H_1(1 + G_1G_2H_2)}$$

12. (A) There cannot be common subscript because subscript refers to node number. If subscript is common, that means that node is in both loop.

13. (D) $L_1 = -bc$, $L_2 = -fg$, $L_3 = jgic$, $L_1L_3 = bcfg$

$$\Delta = 1 - (-bc - fg + cigj) + bcfg = 1 + bc + fg - cigj + bcfg$$

14. (A) In this graph there are three feedback loop. $abef$ is not a feedback path because between path x_2 is a summing node.

15. (B) By putting $R(s) = 0$

$$P_1 = -H_2G_1, L_1 = -G_1H_2H_1, \Delta_1 = 1, \quad T_n(s) = \frac{-H_2G_1}{1 + G_1H_2H_1}$$

if $|G_1H_2H_1| \gg 1$, $T_n(s) = \frac{-H_2G_1}{G_1H_2H_1} = \frac{-1}{H_1}$

16. (C) $P_1 = G$, $L_1 = -H_1$, $L_2 = -H_2$, $L_1L_2 = H_1H_2$, $\Delta_1 = 1$

$$T(s) = \frac{G}{1 + H_1 + H_2 + H_1H_2} = \frac{G}{(1 + H_1)(1 + H_2)}$$

17. (B) $G_a = 1$, $G_b = 1 + 1 = 2$, $G_c = \frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = 1$

There are no loop in any graph. So option (B) is correct.

18. (B)

P. $P_1 = ab$, $\Delta = 1$, $L = 0$, $T = ab$

Q. $P_1 = a$, $P_2 = b$, $\Delta = 1$, $L = \Delta_k = 0$, $T = a + b$

R. $P_1 = a$, $L_1 = b$, $\Delta = 1 - b$, $\Delta_1 = 1$, $T = \frac{a}{a-b}$

S. $P_1 = a$, $L_1 = ab$, $\Delta = 1 - ab$, $\Delta_1 = 1$, $T = \frac{a}{1-ab}$

19. (A) Between e_1 and e_2 , there are two parallel path. Combining them gives $t_a + t_b$. Between e_2 and e_4 there is a path given by total gain $t_c t_d$. So remove node e_3 and place gain $t_c t_d$ of the branch $e_2 e_4$. Hence option (A) is correct.

20. (A) Option (A) is correct. Best method is to check the signal flow graph. In block diagram there is feedback from 4 to 1 of gain $-H_1H_2$. The signal flow graph of option (A) has feedback from 4 to 1 of gain $-H_1H_2$.

21. (C) Consider the block diagram as SFG. There are two feedback loop $-G_1G_2H_1$ and $-G_2G_3H_2$ and one forward path $G_1G_2G_3$. So (D) is correct option.

22. (B) Consider the block diagram as a SFG. Two forward path G_1G_2 and G_3 and three loops $-G_1G_2H_2$, $-G_2H_1$, $-G_3H_2$. There are no nontouching loop. So (B) is correct.

23. (C) $P_1 = 5 \times 3 \times 2 = 30$, $\Delta = 1 - (3 \times -3) = 10$

$$\Delta_1 = 1, \quad \frac{C}{R} = \frac{30}{10} = 3$$

24. (A) $P_1 = 2 \times 3 \times 4 = 24$, $P_2 = 1 \times 5 \times 1 = 5$

$$L_1 = -2, \quad L_2 = -3, \quad L_3 = -4, \quad L_4 = -5, \\ L_1L_3 = 8, \quad \Delta = 1 - (-2 - 3 - 4 - 5) + 8 = 23,$$

$$\Delta_1 = 1, \quad \Delta_2 = 1 - (-3) = 4,$$

$$\frac{C}{R} = \frac{24 + 5 \times 4}{24} = \frac{44}{23}$$

25. (B) $P_1 = G_1G_2$, $P_2 = G_3G_2$

$$L_1 = -G_3G_2H_1, \quad L_2 = -G_1G_2H_1, \quad L_3 = G_4, \quad \Delta_1 = \Delta_2 = 1$$

There are no nontouching loop.

$$T(s) = \frac{P_1\Delta_1 + P_2\Delta_2}{1 - (L_1 + L_2 + L_3)} = \frac{G_1G_2 + G_2G_3}{1 + G_1G_2H_1 + G_2G_3H_1 - G_4}$$

26. (C) $P_1 = G_1G_2$, $L_1 = -G_1G_2H_1H_2$, $L_2 = G_2H_2$

$$\frac{C(s)}{R(s)} = \frac{G_1G_2}{1 + G_1G_2H_1H_2 - G_2H_2}$$

27. (B) There is one forward path G_1G_2 .

Four loops $-G_1G_4$, $-G_1G_2G_8$, $-G_1G_2G_5G_7$ and $-G_1G_2G_3G_6G_7$.

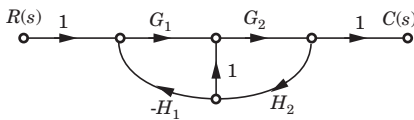


Fig. S6.1.27

There is no nontouching loop. So (B) is correct.

28. (A) SFG:

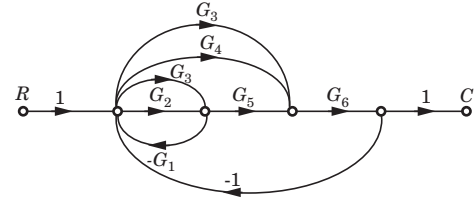


Fig. S6.1.28

$$P_1 = G_2G_5G_6, \quad P_2 = G_3G_5G_6, \quad P_3 = G_3G_6, \quad P_4 = G_4G_6$$

If any path is deleted, there would not be any loop.

$$\text{Hence } \Delta_1 = \Delta_2 = \Delta_3 = \Delta_4 = 1$$

$$\frac{C}{R} = \frac{G_4G_6 + G_3G_6 + G_3G_5G_6 + G_2G_5G_6}{\Delta}$$

29. (A)

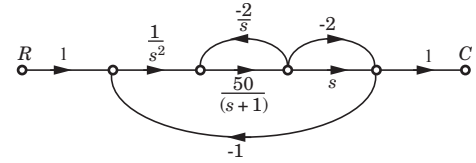


Fig. S6.1.29

$$P_1 = \frac{1}{s^2} \cdot \frac{50}{(s+1)} \cdot s = \frac{50}{s(s+1)}$$

$$P_2 = \frac{1}{s^2} \cdot \frac{50}{s+1} \cdot (-2) = \frac{-100}{s^2(s+1)}$$

$$L_1 = \frac{50}{s+1} \cdot \frac{-2}{s} = \frac{-100}{s(s+1)}$$

$$L_2 = \frac{1}{s^2} \cdot \frac{50}{s+1} \cdot s \cdot (-1) = \frac{-50}{s(s+1)}$$

$$L_3 = \frac{1}{s^2} \cdot \frac{50}{s+1} \cdot (-2) \cdot (-1) = \frac{100}{s^2(s+1)}$$

$$\Delta = 1 + \frac{100}{s(s+1)} + \frac{50}{s(s+1)} - \frac{100}{s^2(s+1)}$$

$$\Delta_1 = \Delta_2 = 1$$

$$\frac{C}{R} = \frac{P_1 + P_2}{\Delta} = \frac{50(s-2)}{s^3 + s^2 + 150s - 100}$$

30. (D) $P_1 = G_1G_2G_3$

$$L_1 = -G_1H_1, \quad L_2 = -G_2H_2, \quad L_3 = -G_3H_3$$

$$L_1L_3 = G_1G_3H_1H_3$$

$$\Delta = 1 - (-G_1H_1 - G_2H_2 - G_3H_3) + G_1G_3H_1H_3$$

$$\Delta = 1 + G_1H_1 + G_2H_2 + G_3H_3 + G_1G_3H_1H_3$$

$$\Delta_1 = 1$$

$$\frac{C}{R} = \frac{G_1G_2G_3}{1 + G_1H_1 + G_2H_2 + G_3H_3 + G_1G_3H_1H_3}$$